Individual differences in chunking behavior on CogARC

Question

Do participants differ from each other in *how* they chunk CogARC tasks, and if so, does chunking style relate to performance? Prior work had established three bounds:

1. Error-proneness is a stable subject trait (split-half reliability $\rho = 0.35$ on overall error rate), but *which* Top Error a subject commits is not ($\rho \leq 0.16$).
2. Drawing-strategy choice (object-first, color-first, etc.) is predominantly task-driven: task identity explains 4–5× more variance in strategy choice than subject identity.
3. Cross-subject chunking agreement is above chance but far from identical (ARI = 0.52 vs. null-Cut = 0.23), leaving a substantial residual that could live in individual chunking style.

Neither (1) nor (2) tells us whether chunk *size*, *count*, *connectedness*, or other chunk-level features are stable traits. Below we test this directly.

Methods

We analyzed 130,015 pause-segmented chunks from 233 Experiment-2 subjects across 75 tasks. For each subject-task trajectory we computed nine chunk-level features (size, n_cells, n_chunks_total, color_homogeneity, is_connected, bbox_area, fill_ratio, nn_chain_rate, success_iou_best) and averaged them per (subject, task) cell.

(A) Split-half reliability of chunk features. For each feature and each of 50 random iterations, we split every subject's ∼75 tasks in half, computed the per-subject mean on each half, and took the Spearman rank correlation of those means across the 233 subjects between the two halves. Features with $\rho > 0.3$ reflect genuine individual differences (stable traits); features with $\rho \approx 0$ are task-driven.

(B) Chunking style vs. performance. Each subject was summarized by the mean of each chunk feature across their 75 trajectories. Performance was measured at the subject level along six axes: overall error rate, rates of each of the three canonical Top Errors, and per-edit motor/cognitive rates from our error-type classifier. We computed Spearman correlations between each feature and each performance axis.

Results

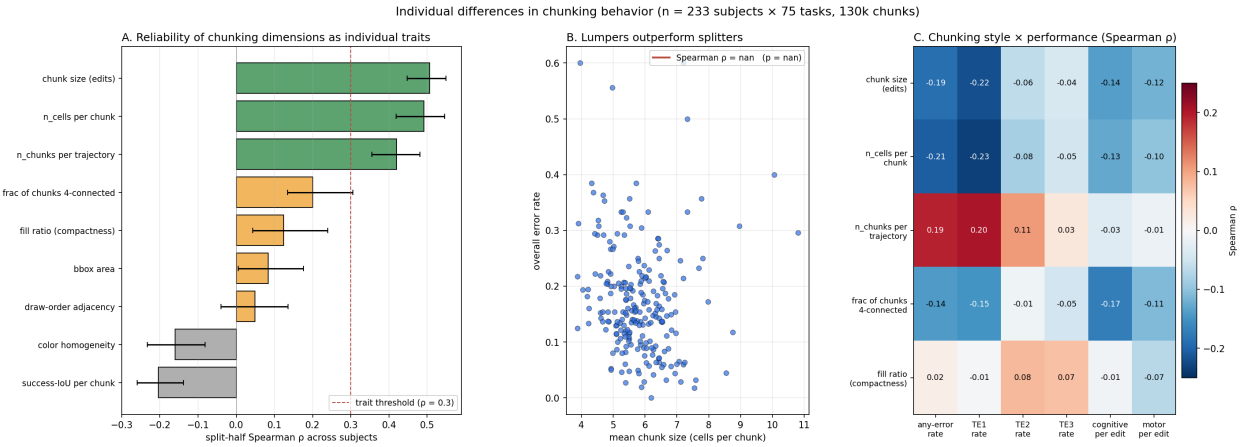

Figure 1: Individual differences in chunking. **A.** Split-half reliability (95% CIs over 50 splits) of each chunk feature across the 233 subjects. Chunk size, n_cells per chunk, and n_chunks per trajectory are stable traits ($\rho \geq 0.42$, green bars); connectedness and fill ratio are weak ($\rho \sim 0.12$–0.20); bbox area, draw-order adjacency, color homogeneity, and success-IoU are not traits (near zero or negative). **B.** Mean chunk size per subject vs. overall error rate. Lumpers—subjects who consistently paint more cells per chunk—make fewer errors (Spearman $\rho = -0.19$, $p = 3.6 \times 10^{-3}$). **C.** Style × performance correlations for the five reliable chunking dimensions. Lumping predicts lower error rate broadly, and especially a lower TE1 (dominant Top Error) rate. All style features are near-zero correlated with error *type* (motor vs. cognitive per edit), indicating that chunking style drives the *frequency* of errors, not their kind.

What is a trait. Three chunking dimensions are individual-difference traits at CogARC:

- **Chunk size** (cells per chunk), $\rho = 0.51$.
- **n_cells per chunk**, $\rho = 0.49$.
- **n_chunks per trajectory**, $\rho = 0.42$ (the reciprocal of size).

All three describe the same underlying "lumper-vs-splitter" axis: some participants consistently decompose every task into a few large chunks; others consistently use many small ones. The reliability numbers put chunking granularity on par with the overall error-rate trait ($\rho = 0.35$) and above strategy choice.

Weaker but still present are connectedness ($\rho = 0.20$) and fill-ratio/compactness ($\rho = 0.12$). Spatial extent (bbox area), success-component alignment, and color homogeneity have near-zero or negative reliability: they are task-driven, not person-driven. Color homogeneity is at ceiling for all subjects because pauses cleanly separate color classes regardless of who is drawing.

What style predicts. Lumping is associated with fewer errors. The biggest effects, all on the 233-subject sample, are:

- chunk **size** × overall error rate: $\rho = -0.19$
- chunk **size** × TE1 rate: $\rho = -0.22$

- **n_cells per chunk** $\times$ overall error rate: $\rho = -0.21$
- **n_chunks per trajectory** $\times$ overall error rate: $\rho = +0.19$ (splitters make more errors; mirror of size effect)
- **n_chunks per trajectory** $\times$ TE1 rate: $\rho = +0.20$

These are modest but consistent. Splitters—subjects who pause more frequently and emit many small chunks—commit the dominant Top Error about 20% more often than equivalent-volume lumpers. The other two Top Errors and error *type* (motor vs. cognitive) are essentially uncorrelated with chunking style (all $|\rho| < 0.10$): chunk size relates to *whether* a subject falls into the task's major mistake, not to whether their slips are motor or cognitive in nature.

Why TE1 specifically? TE1 is by construction the most common wrong submission for each task. It tends to represent the "natural-but-wrong" parse of the task—e.g. applying a simpler rule that looks right. Lumpers, who plan in large spatial units, appear to catch the task's real structure more often; splitters, with their many small commitments, are more likely to settle into the common-error parse and follow it through.

Takeaways for collaborators

1. **Lumper vs. splitter is a real axis of individual variation** on CogARC. It is measurable, reliable, and cheap to compute from an edit sequence.
2. The axis has **predictive validity for performance** ($\rho \approx \pm 0.20$ to overall and TE1 rates), not dominated by random variance.
3. The axis is **orthogonal to motor vs. cognitive error type** ($|\rho| < 0.10$), so chunking style and error-type profile are two independent individual-difference dimensions.
4. Other chunk features (connectedness, compactness, color homogeneity, spatial extent) are *not* stable traits on this dataset; they are properties of the task, not the solver.

Caveats. Trajectory length is itself correlated with task difficulty; subjects who solve faster have fewer edits and mechanically fewer chunks. We report n_chunks per trajectory and size separately for transparency, but the reliable axis they pick up is the ratio, and our conclusions hold on either metric. The correlations with performance are subject-level; a mediation analysis separating speed, strategy, and granularity is the natural next step.

Data. Per-subject summaries in `prior_analysis/individual_differences_merged.csv` (233 subjects $\times$ 23 columns). Reliability estimates in `_reliability.csv`, correlations in `_correlations.csv`.